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CSIT 321-G01

1. Pain Point

- Many online sellers operate across multiple e-commerce platforms such as Shopee, Lazada, and TikTok Shop. However, these platforms do not synchronize inventory automatically, forcing sellers to manually update stock levels across different systems. This often leads to inventory inaccuracies, overselling, order cancellations, and poor customer satisfaction.

2. Solution

- We propose a Centralized Multi-Platform E-Commerce Management System that enables sellers to manage products, orders, inventory, payments, and shipments within a single platform. The system helps streamline operations, reduce inventory inaccuracies, and improve coordination between sellers and customers by integrating essential e-commerce management functions into one system.
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3. For this solution, identify the following:

3.1 Target Users

- Online Sellers / Small Business Owners
 - Sellers operating on multiple e-commerce platforms
 - Need a centralized system to manage products, orders, and inventory
 - Customers
 - Buyers who purchase products online
 - Benefit from accurate stock availability and smoother order processing
 - System Administrators
 - Manage the system database and monitor transactions
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3.2 Must-have Requirements (Functional Requirements)

Core features required for the system to function.

- Manage products and categories
 - Track and update inventory levels
 - Record orders and order items
 - Identify the platform where each order originates
 - Process payments
 - Monitor shipment and delivery status
 - Automatically update inventory when orders occur
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3.3 Should-have Requirements (Non-functional Requirements)

Important features that improve the system but are not critical.

- Allow customers to submit product reviews
 - Provide basic reporting for sellers
 - Display order history and tracking
 - Support multiple sellers using the system
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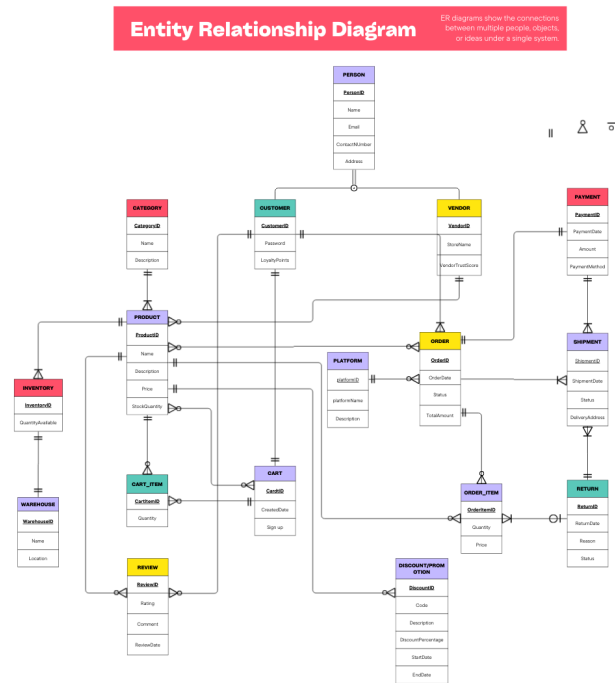
3.4 Nice-to-have Requirements (Dashboard of the Solution)

Features that are outside the scope of the current system.

- Integration with real marketplace APIs
- Advanced AI product recommendations
- Real-time live logistics tracking
- Full mobile application

4. Based on the items listed in 3.2, 3.3 and 3.4, draw the following diagrams:

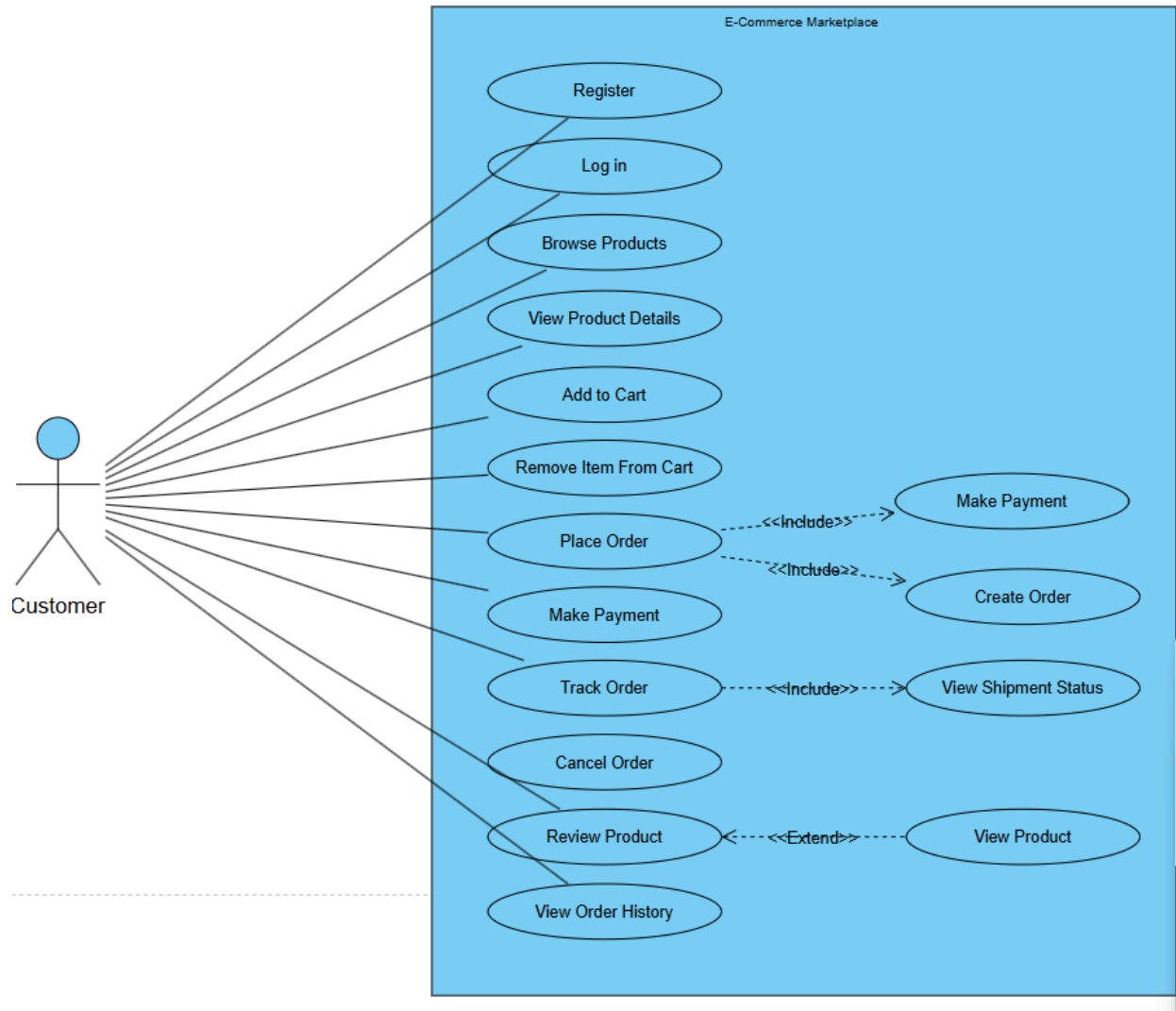
4.1 Entity Relationship Diagram (no FK in the ERD)

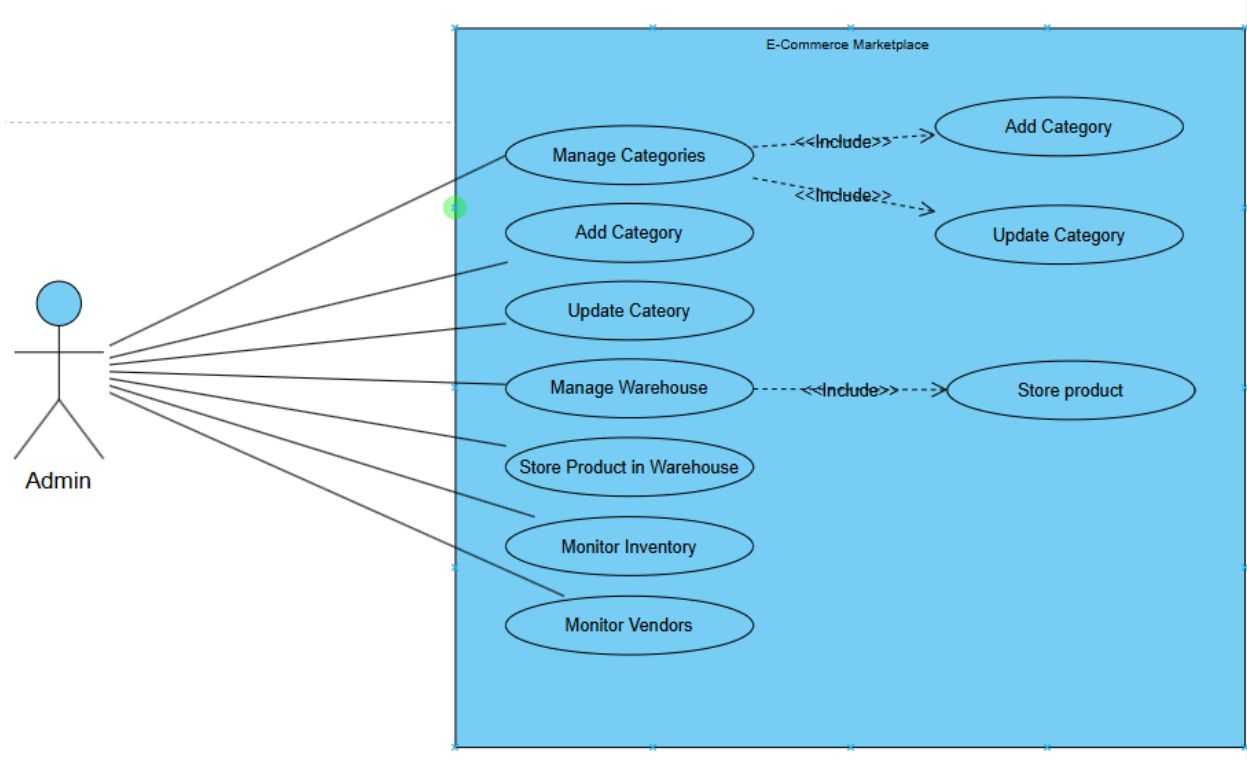
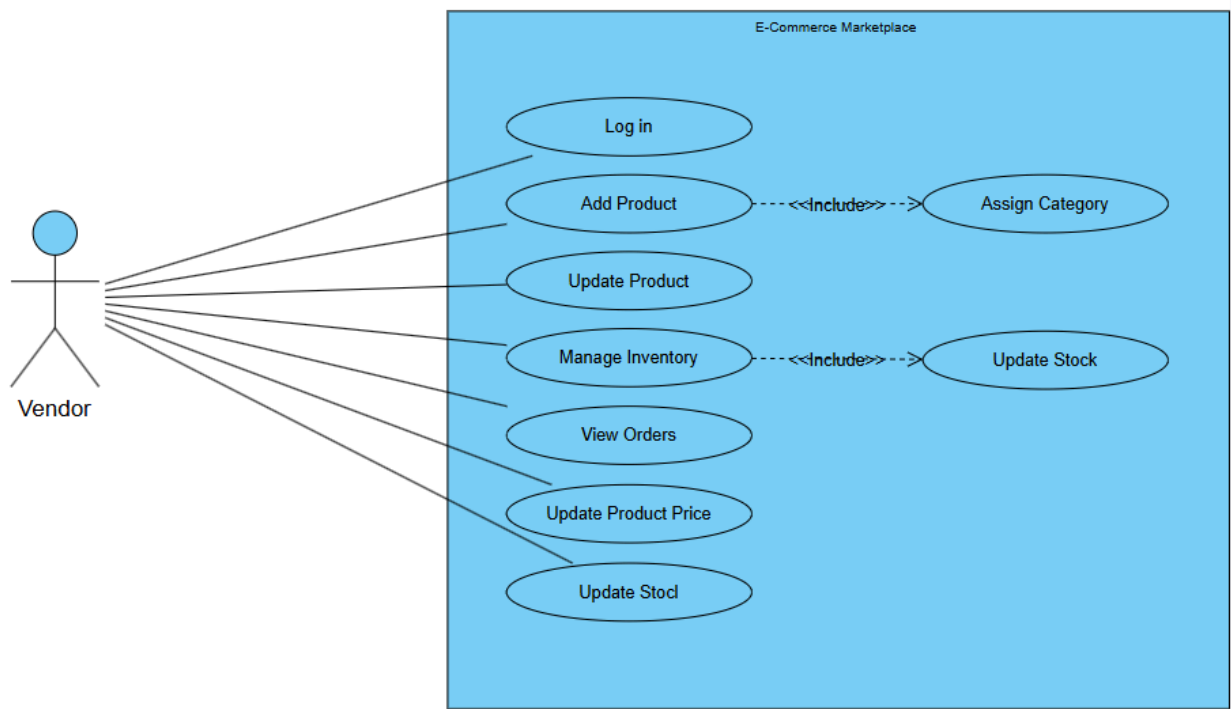


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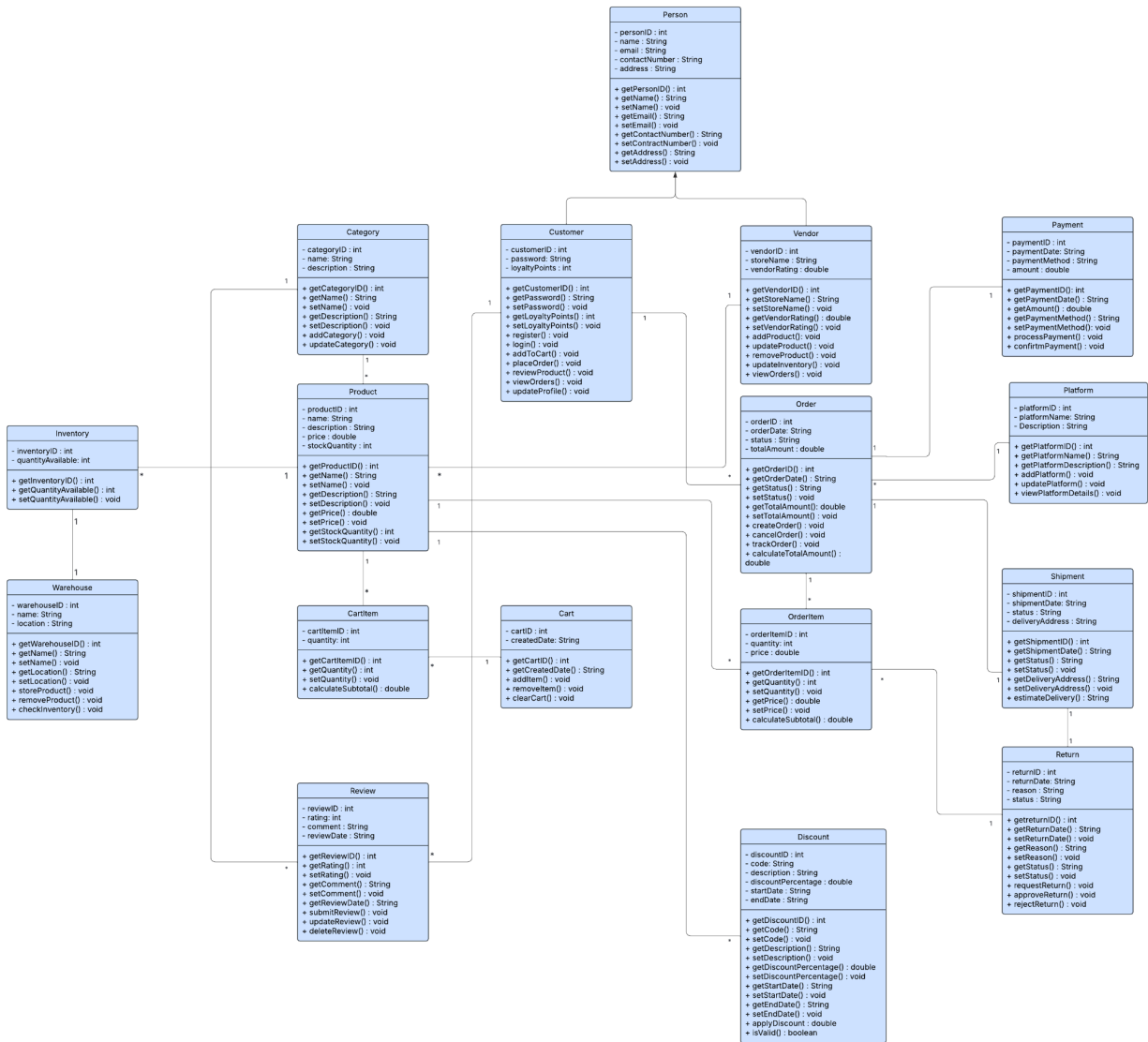
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4.2.) Use Case Diagram





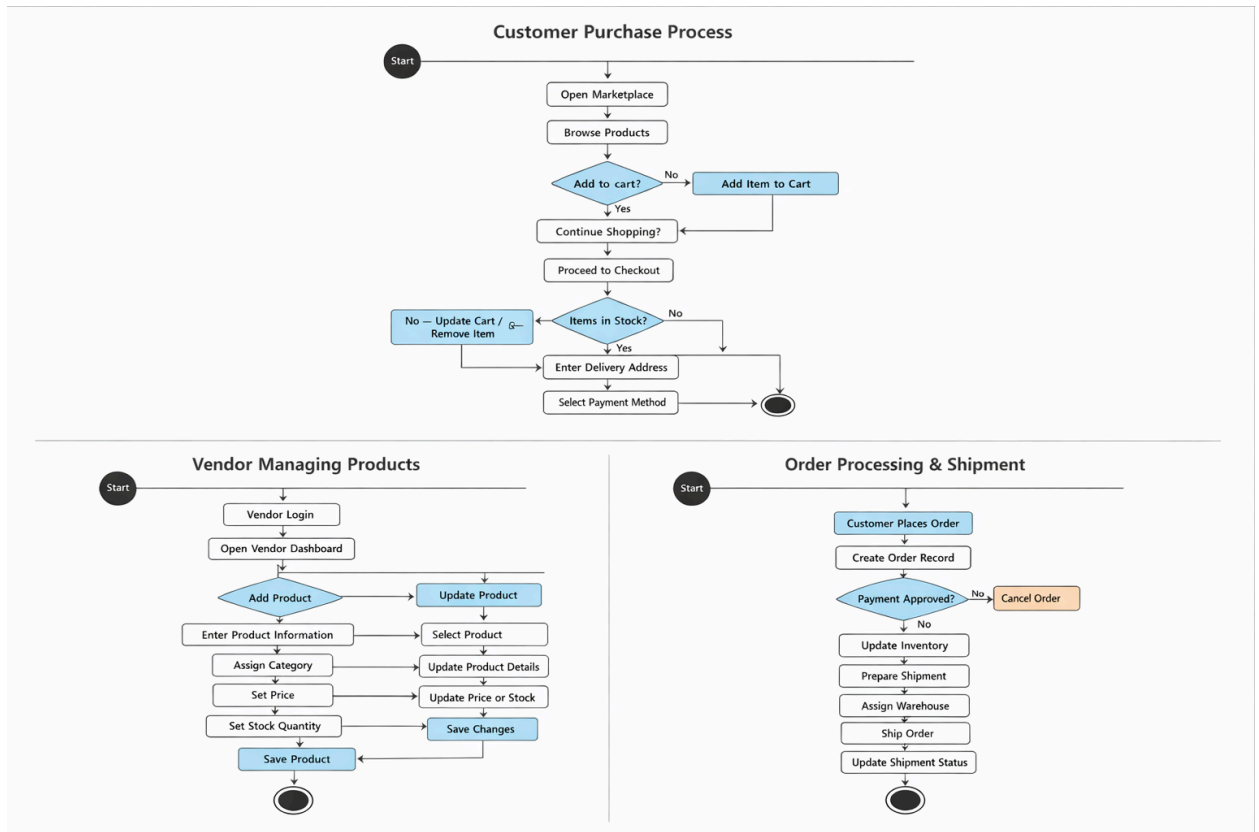
4.3.) Class Diagram



LucidChart Link:

https://lucid.app/lucidchart/094f63df-249a-40ee-a147-5d4e53fc65c5/edit?beaconFlowId=6BFC62024BE2EC34&invitationId=inv_11383743-7665-4967-9968-f4ff51cf38ef&page=HWEp-vi-RSF0#

4.4.) Activity Diagram for each use case in the Use Case Diagram



4.2.) Figma Prototype (with transitions)

<https://www.figma.com/proto/bqud3QQ20hdNuifTEzF9xS/Untitled?node-id=0-1&t=cCkYRc6BbHEBnREq-1>